

Dividing up the unrestricted partitions

OPG-37222

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Begin with the generating function for unrestricted partitions:

$$(1+x+x^2+\dots)(1+x^2+x^4+\dots)(1+x^3+x^6+\dots)\dots$$

Now change some of the plus signs to minus signs. The resulting series will have coefficients congruent, mod 2, to the coefficients of the generating series for unrestricted partitions. I conjecture that the signs may be chosen such that all the coefficients of the series are either 1, -1, or zero.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>